Quantum Random Walks for Market Microstructure

Phase 6 final presentation BTCUSDT, June 12, 2026

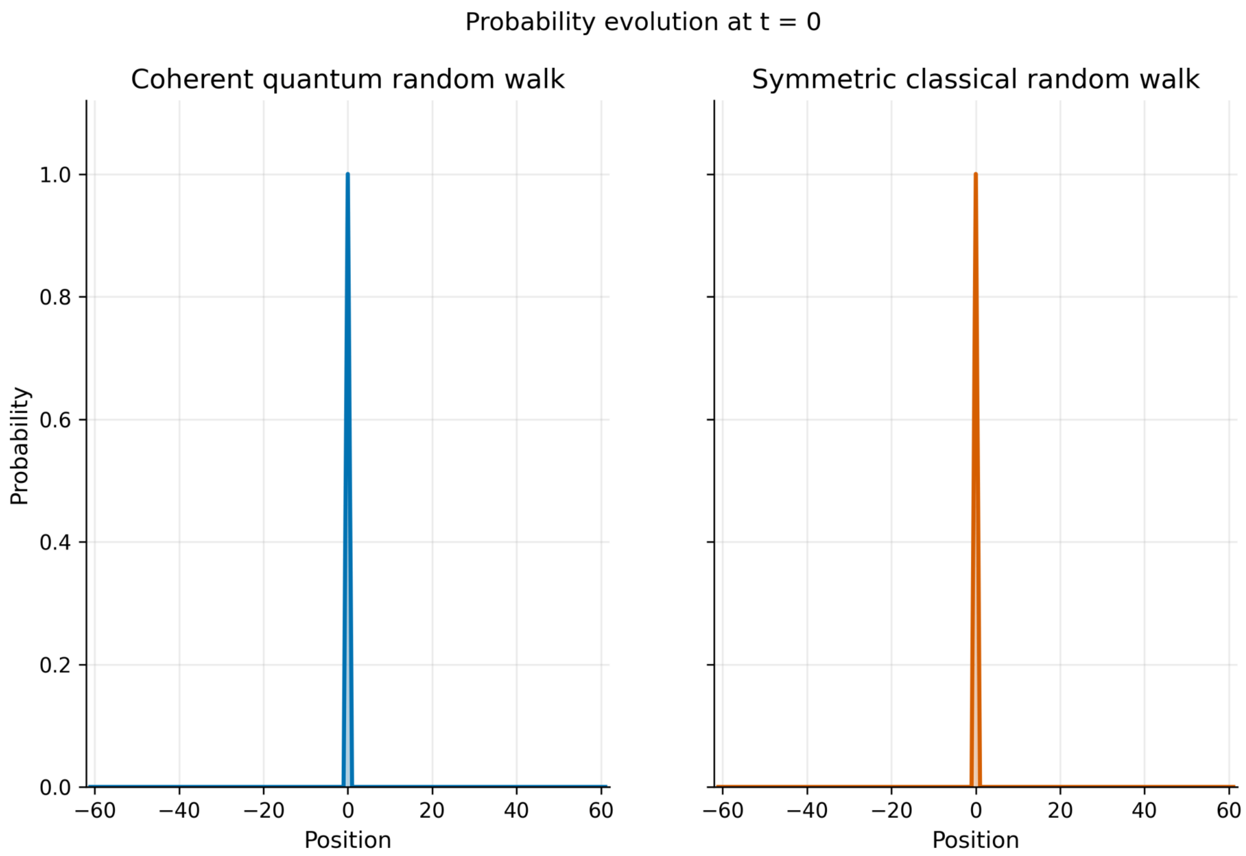
Research Question

Can a causally calibrated QRW reproduce short-horizon market properties better than classical baselines?

Why QRW?

Interference, ballistic spreading, and a tunable coherent-to-diffusive transition.

QRW vs CRW



The coherent walk spreads with interference peaks; the classical walk follows a diffusive binomial envelope.

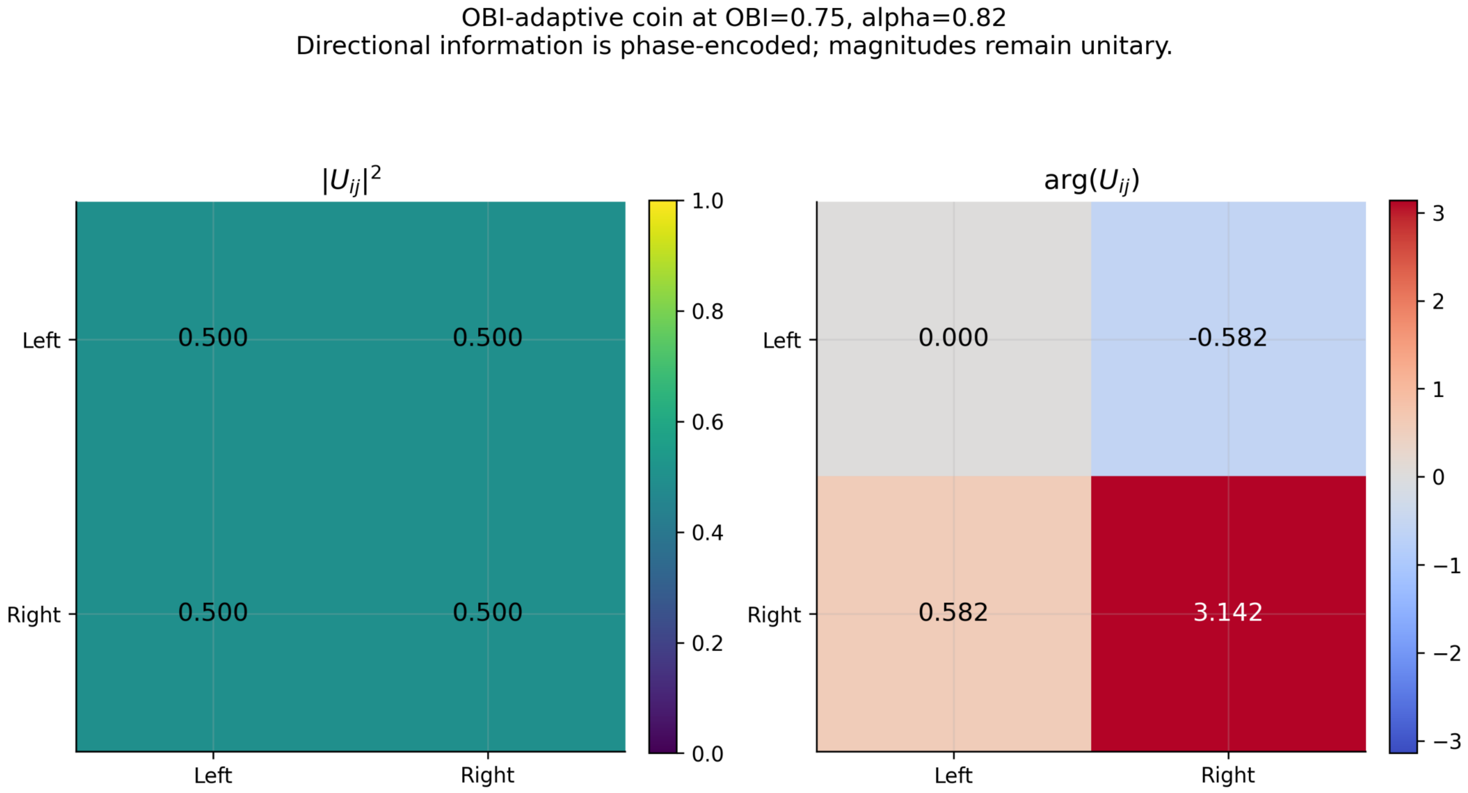
Active Data

1,869,214 training rows 1,246,144 later holdout rows Historical June 1-7 derived artifacts remain invalidated.

Causal Pipeline

Trailing cleaner -> chronological split -> disjoint validation -> fixed seed -> matched statistical samples

Market Mapping



OBI and direction enter a unitary phase-adaptive coin. Trade intensity controls basis dephasing.

Benchmark Models

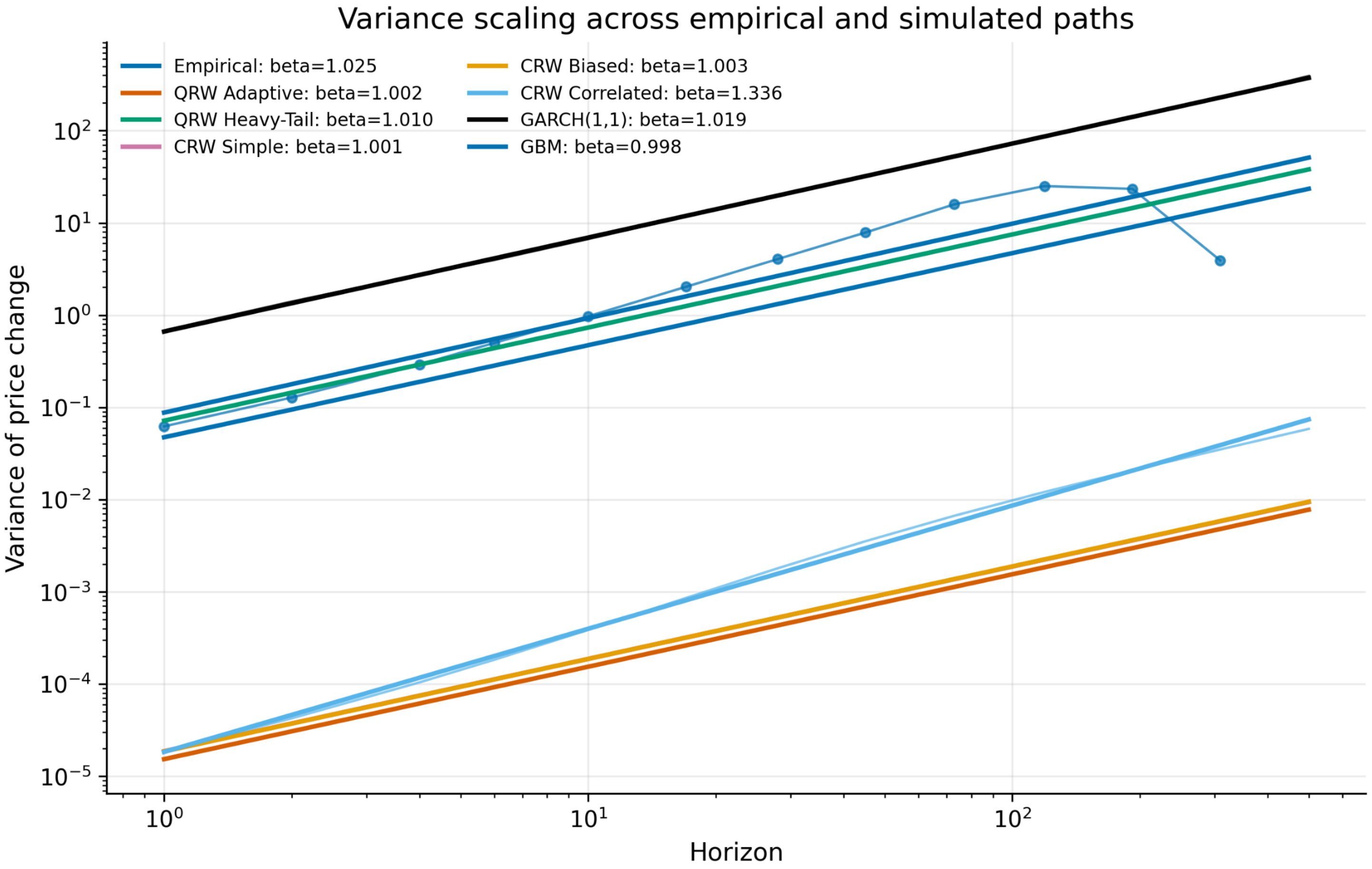
QRW Adaptive CRW Simple / Biased / Correlated GARCH(1,1) GBM

Evaluation

Distribution tests Variance scaling Autocorrelation Tail risk

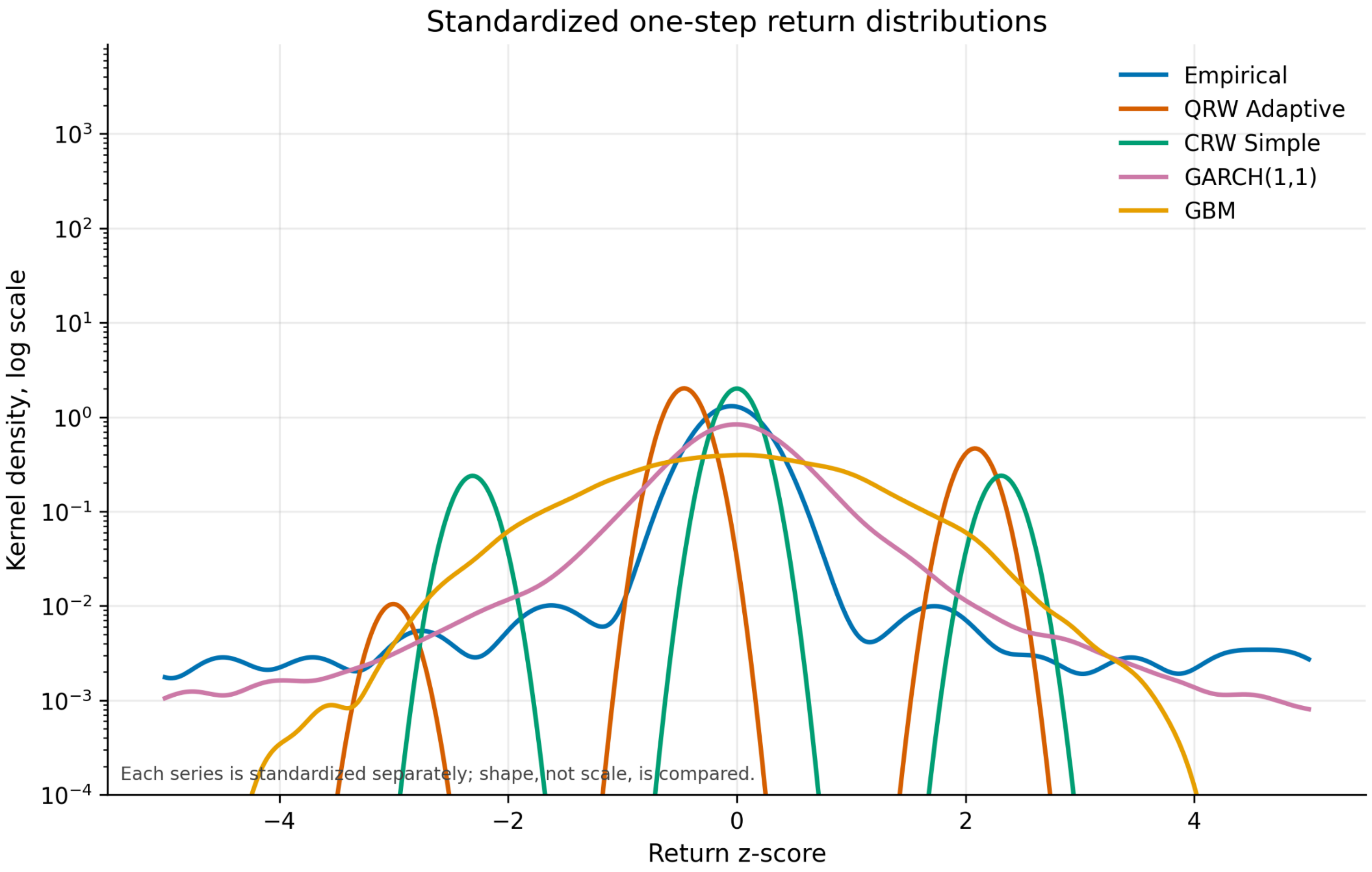
Eight-metric scorecard

Variance Scaling



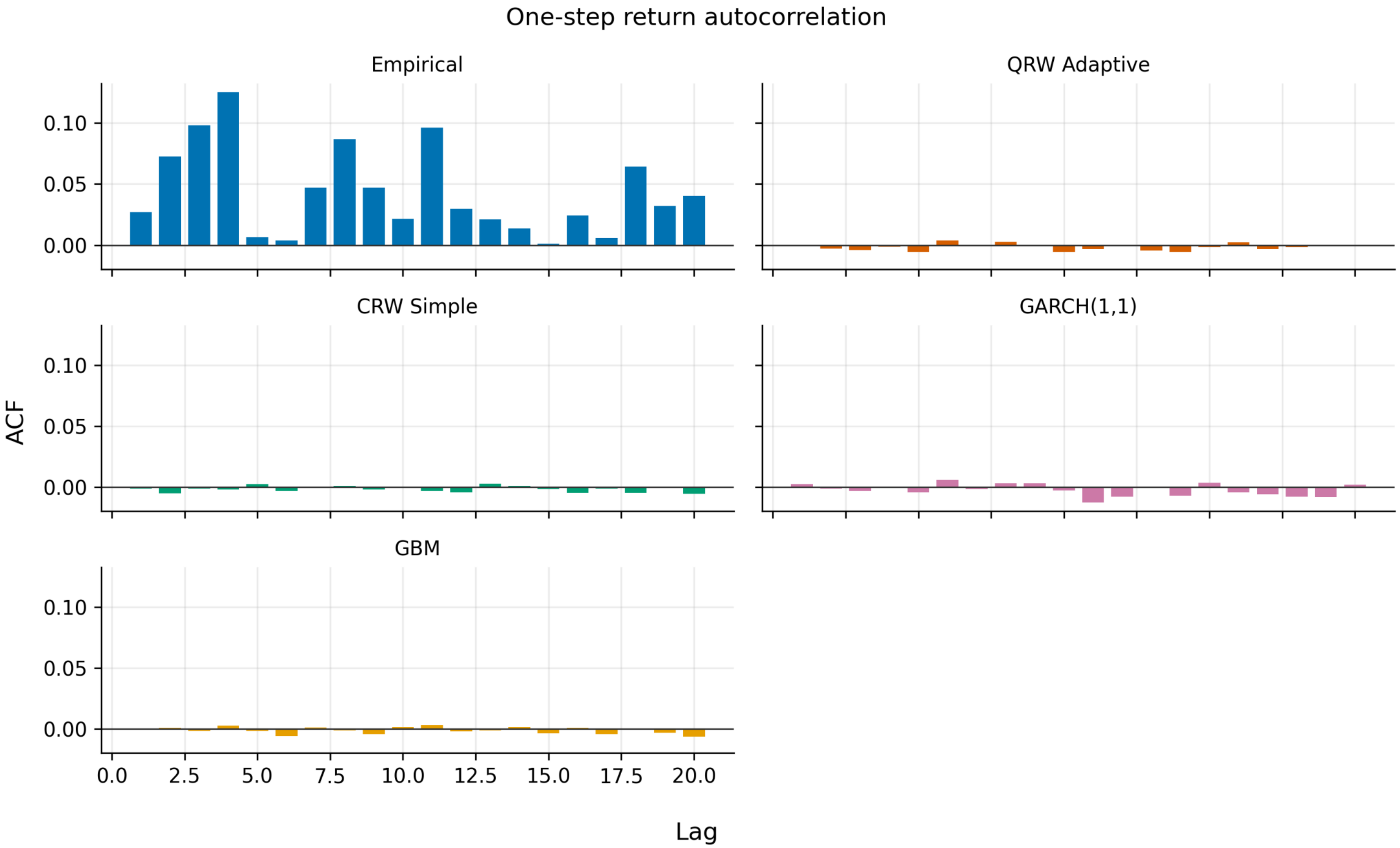
QRW beta = 1.0026 [0.9841, 1.0183]

Distribution Shape



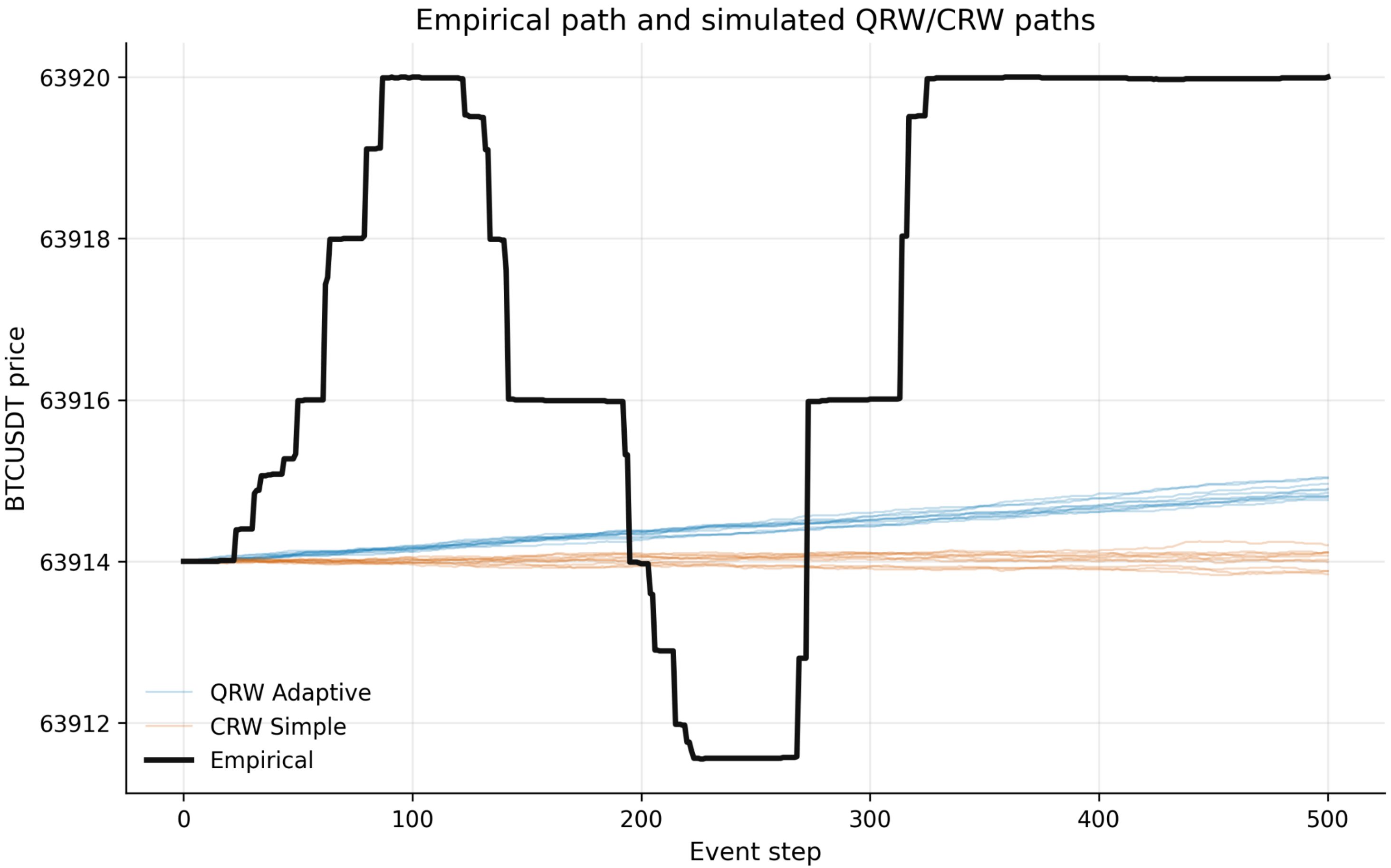
QRW adjusted KS p-value = 9.553e-04

Autocorrelation



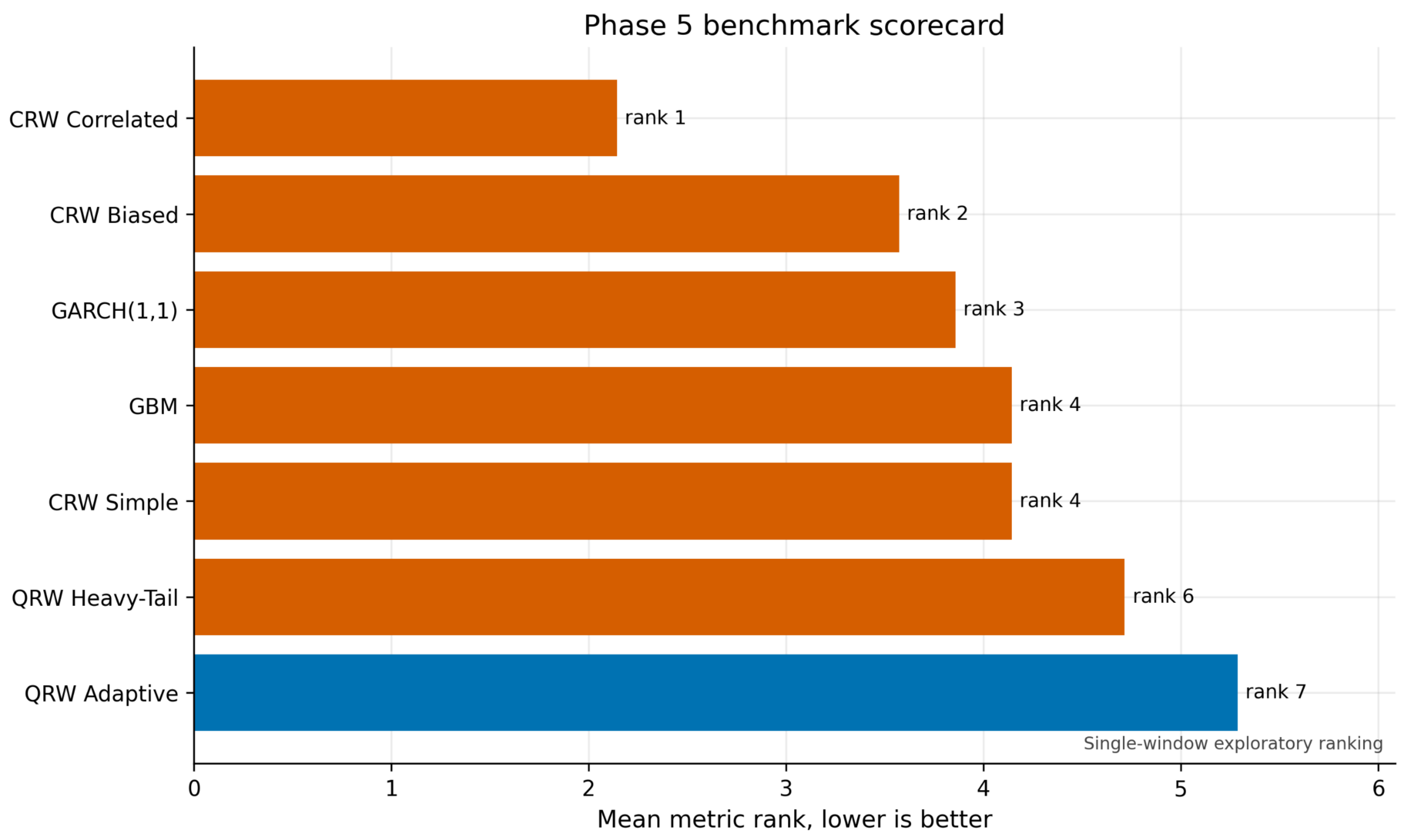
QRW ACF MSE = 0.003275

Sample Paths



Path overlays expose scale and directional differences hidden by aggregate ranks.

Scorecard



CRW Correlated ranks first. QRW Adaptive ranks 7.

What We Can Claim

The engineering and reporting pipeline passes. Current evidence does not establish QRW predictive or distributional superiority.

Next Confirmatory Study

Freeze the protocol, collect at least 20 fresh UTC days with synchronized LOB data, then execute one untouched evaluation.